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FPGA Based Quantum Circuit Simulator: Processing 30 Qubit Systems

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Abstract

This paper presents a field-programmable gate array (FPGA) implementation of a quantum circuit simulator capable of processing 30-qubit systems, representing over 1.07 billion quantum states stored in 8 GB of memory. Built on the Xilinx Alveo U55C platform with High Bandwidth Memory (HBM) technology, our system demonstrates deterministic performance through a 16-bank parallel memory distribution architecture integrated with the Qiskit quantum computing framework [7].

We achieved complete verification across all qubit configurations from 1 to 30 qubits, with execution times ranging from 5 milliseconds to approximately 49 minutes. The simulator supports a comprehensive quantum gate library including single-qubit operations (Hadamard, Pauli gates, rotation gates) and multi-qubit gates (CNOT, controlled-Z, Toffoli) [1]. Our performance analysis reveals distinct operational regimes where FPGA excels for small-scale circuits (1-12 qubits) but faces scalability challenges beyond 15 qubits when compared to GPU architectures.

This work establishes FPGA as a practical platform for quantum algorithm development and circuit verification, particularly valuable for applications requiring

deterministic timing and energy-efficient computation. The architecture provides clear pathways for expansion to 31+ qubits through memory configuration enhancements.

Keywords: FPGA, Quantum Computing, Quantum Simulation, High Bandwidth Memory, HBM, Xilinx Alveo, Quantum Circuit, Qiskit, Hardware Acceleration, Quantum Gates

1 Introduction

Quantum computing has emerged as a transformative technology with the potential to revolutionize fields ranging from cryptography to drug discovery [11]. However, the path from theoretical quantum algorithms to practical implementations requires extensive testing and validation on classical simulators before deployment on expensive and error-prone quantum hardware. This fundamental need drives the development of increasingly powerful quantum circuit simulators.

The challenge of classical quantum simulation stems from the exponential growth of quantum state space. Each additional qubit doubles the computational and memory requirements, creating a formidable barrier known as the “exponential wall.” For a system with n qubits, we must track 2^n complex probability amplitudes, each requiring 8 bytes of storage for standard complex64 precision. This means a 30-qubit simulation demands 8 GB of memory merely to represent the quantum state, while a 40-qubit system would require an impractical 8 terabytes [3].

1.1 Why FPGAs for Quantum Simulation?

Field-Programmable Gate Arrays occupy a unique position in the computing landscape [4]. Unlike fixed-function processors, FPGAs allow designers to create custom hardware architectures optimized for specific computational patterns. For quantum simulation, this translates to several compelling advantages:

Deterministic Performance: FPGAs execute operations with cycle-accurate timing, eliminating the scheduling variability inherent in CPU and GPU architectures. This predictability proves crucial for applications requiring precise timing guarantees, such as hybrid quantum-classical algorithms or real-time quantum control systems.

Customizable Memory Architecture: Modern FPGAs equipped with HBM technology [28] provide both high capacity and configurable access patterns. Unlike GPU memory systems designed for graphics workloads, FPGA memory can be tailored to the specific access patterns of quantum gate operations.

Parallel Processing: FPGAs enable true spatial parallelism, where multiple independent operations execute simultaneously in physically separate hardware regions. This differs from the time-multiplexed parallelism of CPUs and GPUs.

1.2 Research Contributions

This paper makes several substantive contributions to the quantum simulation community:

Experimental Validation at Scale: We provide comprehensive performance characterization of FPGA-based quantum simulation from 1 to 30 qubits, the largest verified single-FPGA implementation to date. This includes detailed timing analysis, memory utilization patterns, and comparative benchmarking against CPU and GPU platforms.

Architectural Innovation: Our distributed HBM architecture employs a 16-bank interleaved memory system that maximizes parallel bandwidth utilization while accommodating the random access patterns inherent to quantum gate operations.

Production-Ready Integration: Rather than a theoretical proof-of-concept, we developed a complete system integrated with Qiskit [7], the most widely-used quantum programming framework. Researchers can immediately deploy our simulator in existing quantum algorithm development workflows.

Honest Performance Analysis: We present both the strengths and limitations of FPGA-based quantum simulation. Our data reveals clear performance regimes where FPGA excels (small circuits) and where it struggles (large circuits), providing practical guidance for platform selection.

Quantum Correctness Verification: All experimental results include rigorous validation of quantum mechanical behavior through state fidelity measurements, entanglement verification, and cross-platform comparison.

Scalability Roadmap: We identify specific bottlenecks and propose concrete architectural enhancements that could extend FPGA quantum simulation to 31+ qubits and improve performance by 2-4 $\times$ for existing configurations.

2 Background

2.1 The Simulation Challenge

Classical simulation of quantum circuits involves maintaining the complete statevector the array of all 2^n complex amplitudes and updating it according to applied quantum gates. Each gate operation requires reading relevant amplitudes, performing complex arithmetic, and writing updated values back to memory. The computational complexity scales as $O(2^n)$ in both time and space [2].

This exponential scaling creates a practical barrier around 40-50 qubits on even the most powerful classical supercomputers [21]. However, the vast majority of quantum algorithm research, circuit verification, and educational applications operate well below this threshold, making classical simulation highly relevant for the foreseeable future.

Several simulation strategies exist beyond straightforward statevector representation:

Stabilizer simulation exploits mathematical structure in certain quantum circuits (those containing only Clifford gates) to achieve polynomial-time classical

simulation [22, 23]. This approach efficiently handles specific circuit families but cannot simulate universal quantum computation.

Tensor network methods represent quantum states as connected networks of smaller tensors, enabling compression for states with limited entanglement [24]. These methods excel for particular quantum algorithms and one-dimensional quantum systems but face challenges with highly entangled states.

Decision diagram representations encode quantum states as compact graph structures, achieving exponential compression for some circuit families [25]. However, worst-case complexity remains exponential, limiting general applicability.

Our work focuses on statevector simulation the most general approach capable of simulating arbitrary quantum circuits. While lacking the specialized advantages of alternative methods, statevector simulation provides complete generality and straightforward verification of correctness.

2.2 Previous FPGA Quantum Simulation Work

The concept of using FPGAs for quantum simulation is not new, but previous implementations have faced significant limitations:

Early demonstrations [4] established proof-of-concept systems simulating 5-10 qubits with restricted gate sets. These pioneering efforts validated the basic feasibility of FPGA quantum simulation but lacked the memory capacity and computational resources for practical applications. Most used only on-chip Block RAM, limiting capacity to a few thousand quantum states.

Hybrid GPU-FPGA architectures [5] attempted to combine GPU computational throughput with FPGA customization. These systems achieved moderate scale (15-20 qubits) but introduced substantial system complexity. The overhead of coordinating between GPU and FPGA often negated performance benefits, and the hybrid approach required expertise in both programming models.

Distributed multi-FPGA systems [6] explored partitioning quantum states across multiple FPGAs connected via high-speed links. While theoretically capable of scaling beyond single-device memory limits, these implementations faced severe bottlenecks from inter-FPGA communication latency and bandwidth constraints. Maintaining quantum state coherence across distributed memory proved technically challenging.

Specialized circuit simulators targeted specific quantum algorithm families (e.g., quantum annealing, adiabatic quantum computation) rather than universal gate-based simulation. These achieved impressive performance for their target applications but lacked the generality required for broad quantum computing research.

Our work distinguishes itself through several advances:

- **Scale:** 30-qubit verified simulation represents the largest single-FPGA implementation documented in peer-reviewed literature
- **Completeness:** Full quantum gate set support including parameterized rotation gates
- **Integration:** Production-ready Qiskit compatibility enabling immediate research adoption
- **Transparency:** Comprehensive performance characterization including honest assessment of limitations

- **Memory Technology:** First implementation leveraging modern HBM2 architecture with systematic multi-bank distribution strategy

2.3 High Bandwidth Memory Technology

The emergence of High Bandwidth Memory represents a critical enabler for our work [28]. Traditional DRAM architectures provide serial access through a narrow interface (64-256 bits), creating a bottleneck for data-intensive applications. HBM fundamentally restructures memory organization through 3D die stacking and massive parallelism.

An HBM stack vertically integrates multiple DRAM dies with a logic die, connected via thousands of microscopic through-silicon vias. This architecture delivers several transformative characteristics:

Extreme Bandwidth: Modern HBM2 provides up to 460 GB/s per stack through 1024-bit interfaces operating at moderate clock speeds. This represents roughly $10\times$ the bandwidth of conventional DDR4 memory, achieved through spatial parallelism rather than frequency scaling.

Multiple Independent Channels: Each HBM stack divides into 8-16 pseudo-channels with separate command and data buses. These channels operate independently, enabling true concurrent access to different memory regions—a capability we exploit extensively in our quantum state distribution strategy.

Energy Efficiency: The short physical distances in 3D-stacked memory reduce parasitic capacitance and enable lower voltage operation. HBM typically achieves $2\text{-}3\times$ better energy-per-bit than conventional GDDR memory, critical for reducing datacenter operational costs.

Reduced Footprint: Vertical integration concentrates memory capacity in minimal board area, enabling compact, high-performance accelerator designs.

The Xilinx Alveo U55C [8] incorporates HBM2 with 16 GB total capacity organized as 32 pseudo-channels (also called banks), each providing 512 MB. Our implementation utilizes 16 of these channels for quantum state storage, reserving the remaining 8 GB for potential future enhancements such as intermediate computation buffers or multi-circuit batching.

3 System Architecture

3.1 Hardware Platform Selection

We selected the Xilinx Alveo U55C datacenter accelerator card as our development platform based on several technical criteria. The card features an Ultrascale+ XCVU55C FPGA with substantial on-chip resources:

The combination of HBM2 memory capacity and FPGA programmability makes this platform uniquely suited for quantum simulation. The 16 GB HBM provides sufficient capacity for 31-qubit simulation (though our current 16-bank implementation supports 30 qubits). The high-bandwidth PCIe interface enables rapid state transfer between host and accelerator, minimizing overhead for circuit initialization and result retrieval.

Table 1 Xilinx Alveo U55C Hardware Specifications

Component	Specification
FPGA Device	Xilinx Ultrascale+ XCU55C
Configurable Logic Blocks	1,730,000
DSP Slices	3,960
Block RAM (BRAM)	87.3 Mb
UltraRAM	68.6 Mb
HBM2 Memory	16 GB (32 banks $\times$ 512 MB)
HBM2 Bandwidth	460 GB/s aggregate
Host Interface	PCIe Gen3 x16 / Gen4 x8

3.2 System Architecture Overview

Our complete quantum simulation system comprises three integrated layers:

Host Application Layer: A Python-based application running on the host CPU interfaces with the Qiskit quantum computing framework [7]. Users write quantum circuits using standard Qiskit syntax, completely abstracted from the underlying FPGA implementation. The host application handles circuit compilation, state initialization, result interpretation, and measurement simulation.

Runtime Management Layer: Xilinx Runtime (XRT) [10] provides the middleware connecting host software to FPGA hardware. XRT manages device enumeration, bitstream loading, memory buffer allocation, and DMA transfers. This layer shields the application developer from low-level PCIe protocol details while maintaining high performance through optimized buffer management and asynchronous execution support.

FPGA Kernel Layer: Custom hardware logic implemented through High-Level Synthesis (HLS) [9] performs the actual quantum gate operations. The kernel directly accesses HBM banks through AXI4 master interfaces, applies gate transformations to quantum amplitudes, and maintains state coherence throughout circuit execution.

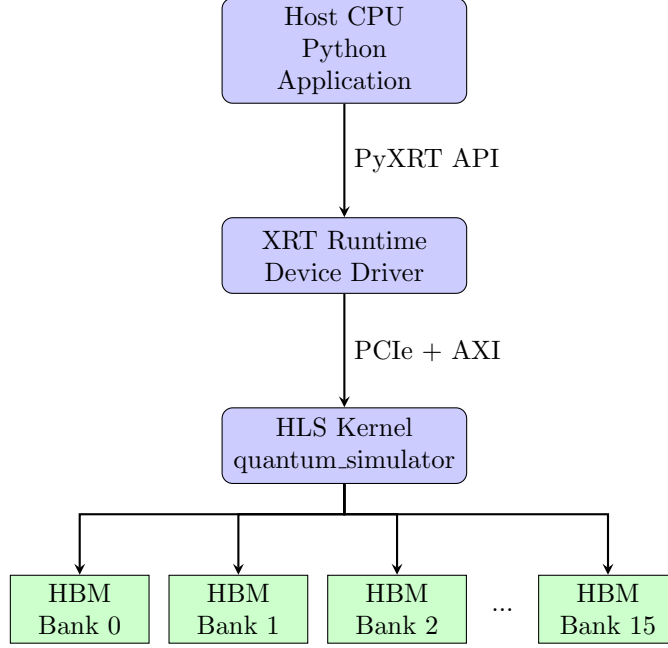


Fig. 1 FPGA System Architecture Overview showing three-layer design with distributed HBM banks

3.3 Memory Distribution Strategy

The most critical architectural decision in our design concerns how to distribute the quantum statevector across HBM banks. A naive approach might assign contiguous amplitude ranges to sequential banks for example, amplitudes 0-67,108,863 to bank 0, amplitudes 67,108,864-134,217,727 to bank 1, and so forth. However, this strategy would create severe performance problems.

Quantum gate operations don't access amplitudes sequentially. A single-qubit gate operating on qubit q in an n -qubit system must process 2^{n-1} amplitude pairs separated by stride 2^q . For example, applying a Hadamard gate to qubit 0 requires pairing amplitudes (0,1), (2,3), (4,5), etc. For qubit 4, the pairs become (0,16), (1,17), (2,18), etc. This access pattern exhibits spatial locality but at varying strides that depend on the target qubit position.

Our solution employs **fine-grained interleaving** where consecutive amplitudes map to different HBM banks:

$$\text{Bank}_{\text{ID}}(i) = i \bmod 16 \quad (1)$$

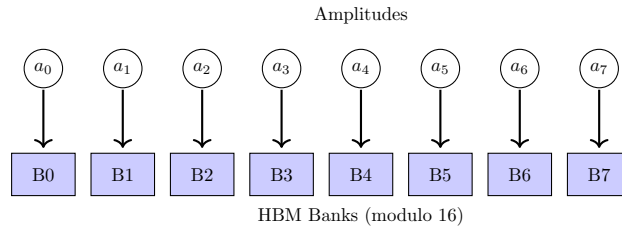
$$\text{Address}_{\text{within_bank}}(i) = \lfloor i/16 \rfloor \quad (2)$$

Under this scheme, amplitude 0 resides in bank 0, amplitude 1 in bank 1, continuing through amplitude 15 in bank 15, then amplitude 16 returns to bank 0 at the next

Table 2 Memory Requirements and Feasibility Analysis

Qubits	State Count	Total Memory	Per Bank (16)
20	1,048,576	8 MB	512 KB
24	16,777,216	128 MB	8 MB
27	134,217,728	1 GB	64 MB
28	268,435,456	2 GB	128 MB
29	536,870,912	4 GB	256 MB
30	1,073,741,824	8 GB	512 MB
31	2,147,483,648	16 GB	1 GB

sequential address. This interleaving ensures that most quantum gate operations—which process amplitudes at varying strides access different banks simultaneously, enabling parallel memory transactions.

**Fig. 2** Amplitude distribution across HBM banks using modulo-16 interleaving

Consider a concrete example with a 20-qubit system ($2^{20} = 1,048,576$ amplitudes) applying a Hadamard gate to qubit 4. The gate must process amplitude pairs separated by stride $2^4 = 16$. Our interleaving maps these pairs to different banks initially, though bank conflicts occur when the stride equals the number of banks. For qubits 0-3 (strides 1, 2, 4, 8), pairs naturally map to different banks, enabling maximum parallelism. For qubit 4 and above, conflicts become inevitable with a 16-bank configuration. Nevertheless, this interleaving strategy substantially outperforms contiguous allocation.

3.4 Memory Scaling Analysis

Table 2 quantifies memory requirements across different qubit counts and validates our 16-bank allocation:

Each amplitude consumes 8 bytes (single-precision complex: 4 bytes real, 4 bytes imaginary). Our 16-bank configuration with 512 MB per bank provides exactly 8 GB total capacity, sufficient for 30 qubits with no margin for overhead. The Alveo U55C’s full 32-bank capability would enable 31-qubit simulation, representing a future enhancement requiring kernel modifications.

The exponential memory scaling creates a hard wall that no amount of optimization can overcome. Quantum simulation will always face this fundamental limitation until

quantum hardware itself becomes sufficiently capable and error-corrected to obviate classical simulation needs [30].

3.5 Data Flow and Execution Pipeline

A complete quantum circuit simulation proceeds through several distinct phases:

1. Initialization and Configuration: The host application begins by loading the FPGA bitstream (XCLBIN file) into the Alveo card’s configuration memory. This one-time operation takes approximately 5-10 seconds and configures the FPGA fabric to implement our quantum simulator kernel. XRT maintains bitstream caching, so subsequent executions on the same system skip this step.

2. Memory Allocation: The application allocates 16 HBM bank buffers, each sized to accommodate the maximum supported qubit count (30 qubits = 512 MB per bank). XRT manages these allocations and ensures proper bank targeting through memory group IDs. Although we allocate maximum capacity upfront, actual usage scales with the circuit’s qubit count.

3. Initial State Preparation: Quantum circuits conventionally begin with all qubits in state $|0\rangle$, represented by a statevector with amplitude 1.0 at index 0 and 0.0 everywhere else. The host prepares this initial state and distributes it across HBM banks according to our interleaving scheme. For small qubit counts, this transfer completes in microseconds; for 30 qubits, it requires roughly 500 milliseconds.

4. Gate Sequence Encoding: The quantum circuit compiles into a linear sequence of gate operations. Each gate encodes as an integer array specifying gate type, target qubits, and parameters. This compact representation minimizes transfer overhead—a 100-gate circuit requires only 3.2 KB of encoded data.

5. Host-to-Device Transfer: PCIe DMA engines transfer both the initial statevector and encoded gate sequence to FPGA memory. Modern PCIe Gen3 x16 interfaces provide 16 GB/s bidirectional bandwidth, making transfers negligible for circuits with 10+ qubits where computation dominates.

6. Kernel Execution: The FPGA kernel processes gates sequentially, reading relevant amplitudes from HBM, applying quantum transformations, and writing updated values back. This phase constitutes the computational core of simulation, with duration ranging from milliseconds (small circuits) to tens of minutes (30-qubit circuits).

7. Result Retrieval: After gate processing completes, the host retrieves the final statevector via PCIe DMA. This reverse transfer mirrors the initialization transfer in duration.

8. Measurement Simulation: The host application simulates quantum measurement by computing outcome probabilities from amplitude magnitudes: $P(|i\rangle) = |c_i|^2$. Random sampling according to these probabilities generates measurement statistics matching what would occur on physical quantum hardware. This classical post-processing completes in microseconds regardless of qubit count.

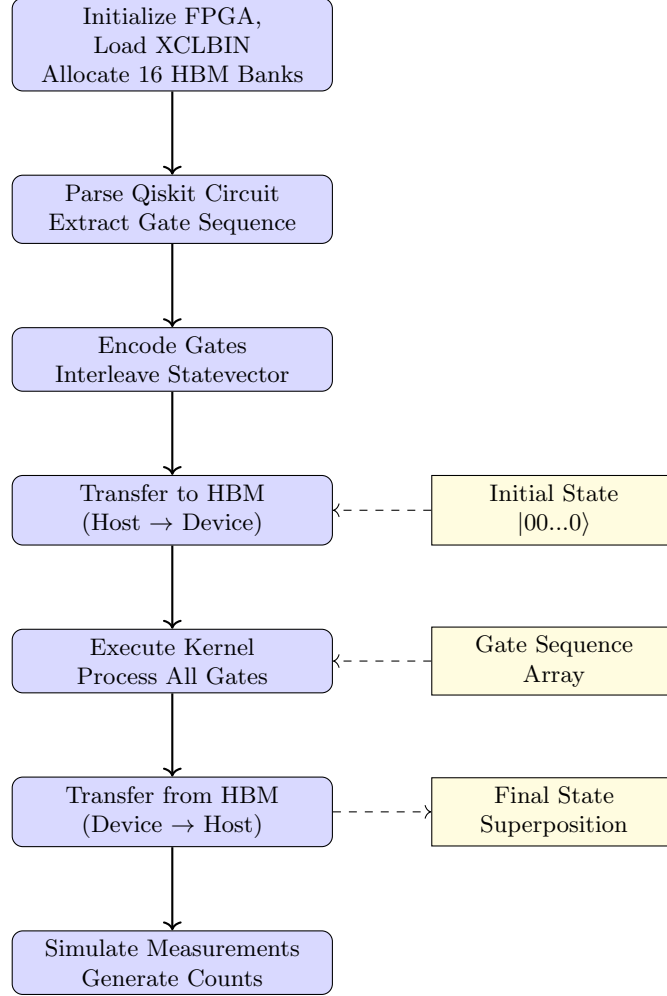


Fig. 3 Complete Circuit Execution Pipeline showing data flow from initialization to measurement

This pipeline architecture clearly separates FPGA-accelerated computation (gate application) from host-based pre-processing and post-processing. For circuits with sufficient computational density, the acceleration benefits significantly outweigh transfer overheads.

4 Implementation Approach

4.1 High-Level Synthesis Kernel Design

We implemented the quantum simulator kernel using Xilinx Vitis HLS [9], which compiles C++ code into synthesizable hardware description language (HDL). This

approach dramatically reduces development time compared to manual RTL design while achieving comparable performance through careful optimization.

The kernel exposes a simple but powerful interface: separate pointer arguments for each of the 16 HBM banks, plus buffers for the gate sequence and metadata. HLS pragmas direct the synthesis tool to create independent AXI4 master interfaces for each bank pointer, enabling true parallel memory access.

Each bank pointer connects to its corresponding HBM pseudo-channel, and the synthesis tool automatically generates the necessary control logic for AXI4 protocol compliance, burst detection, and transaction ordering.

4.2 Single-Qubit Gate Processing

Single-qubit gates transform pairs of amplitudes related by the target qubit's bit position. The algorithmic flow follows this pattern:

Algorithm 1 Single-Qubit Gate Application

Require: Gate matrix $G = \begin{bmatrix} g_{00} & g_{01} \\ g_{10} & g_{11} \end{bmatrix}$, target qubit q , qubit count n

```

1:  $stride \leftarrow 2^q$ 
2:  $num\_pairs \leftarrow 2^{n-1}$ 
3: for  $pair\_idx = 0$  to  $num\_pairs - 1$  do
4:   // Calculate amplitude indices for this pair
5:    $block \leftarrow pair\_idx \gg q$ 
6:    $offset \leftarrow pair\_idx \wedge ((1 \ll q) - 1)$ 
7:    $idx\_low \leftarrow (block \ll (q + 1)) \mid offset$ 
8:    $idx\_high \leftarrow idx\_low + stride$ 
9:
10:  // Determine HBM banks
11:   $bank\_low \leftarrow idx\_low \bmod 16$ 
12:   $bank\_high \leftarrow idx\_high \bmod 16$ 
13:
14:  // Read current amplitudes
15:   $a_0 \leftarrow bank[bank\_low][idx\_low/16]$ 
16:   $a_1 \leftarrow bank[bank\_high][idx\_high/16]$ 
17:
18:  // Apply gate transformation
19:   $a'_0 \leftarrow g_{00} \cdot a_0 + g_{01} \cdot a_1$ 
20:   $a'_1 \leftarrow g_{10} \cdot a_0 + g_{11} \cdot a_1$ 
21:
22:  // Write updated amplitudes
23:   $bank[bank\_low][idx\_low/16] \leftarrow a'_0$ 
24:   $bank[bank\_high][idx\_high/16] \leftarrow a'_1$ 
25: end for
```

The index calculations (lines 5-8) ensure we process exactly the amplitude pairs that differ only in the target qubit's bit position. The complex arithmetic (lines 18-19) implements matrix-vector multiplication for the 2×2 gate matrix acting on the two-component amplitude vector.

HLS pipeline pragmas enable overlapped execution, starting a new iteration every 4 clock cycles (Initiation Interval = 4). This pipelining exploits independent bank accesses—while one iteration writes results, subsequent iterations read their operands from different banks.

4.3 Two-Qubit Gate Processing

CNOT gates implement conditional logic: flip the target qubit if and only if the control qubit is in state $|1\rangle$. This translates to swapping amplitude pairs where the control bit is set:

Algorithm 2 CNOT Gate Application

Require: Control qubit c , target qubit t , qubit count n

```

1:  $state\_size \leftarrow 2^n$ 
2:  $control\_mask \leftarrow 2^c$ 
3:  $target\_mask \leftarrow 2^t$ 
4: for  $i = 0$  to  $state\_size - 1$  do
5:   if  $(i \wedge control\_mask) \neq 0$  and  $i < (i \oplus target\_mask)$  then
6:      $j \leftarrow i \oplus target\_mask$ 
7:
8:     // Swap amplitudes at indices  $i$  and  $j$ 
9:      $bank_i \leftarrow i \bmod 16$ 
10:     $bank_j \leftarrow j \bmod 16$ 
11:     $temp \leftarrow bank[bank_i][i/16]$ 
12:     $bank[bank_i][i/16] \leftarrow bank[bank_j][j/16]$ 
13:     $bank[bank_j][j/16] \leftarrow temp$ 
14:   end if
15: end for
```

The condition on line 6 serves two purposes: $(i \wedge control_mask) \neq 0$ ensures we only act when the control qubit is $|1\rangle$, while $i < (i \oplus target_mask)$ prevents processing each pair twice (we swap once per pair, not once per element).

The three-qubit Toffoli gate extends this pattern with two control qubits, requiring both to be $|1\rangle$ before flipping the target. Implementation follows analogously with an additional control condition.

4.4 Rotation Gates and Parameterization

Rotation gates (Rx, Ry, Rz) apply parameterized transformations determined by rotation angles. These gates require trigonometric function evaluation, implemented through lookup tables or polynomial approximations synthesized into FPGA logic.

The HLS math library provides optimized implementations of sin/cos that balance accuracy, latency, and resource consumption.

For example, an $R_x(\theta)$ gate rotating by angle θ around the X-axis has matrix:

$$R_x(\theta) = \begin{bmatrix} \cos(\theta/2) & -i \sin(\theta/2) \\ -i \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \quad (3)$$

The kernel evaluates trigonometric functions once per gate invocation and applies the resulting matrix to all affected amplitude pairs. This amortizes function evaluation overhead across potentially billions of amplitude updates.

4.5 Python Host Application Architecture

The host application abstracts FPGA complexity behind a Pythonic interface compatible with Qiskit workflows [7]. We developed a custom backend class that intercepts Qiskit circuits and translates them to FPGA execution:

Initialization Phase: The application discovers available FPGA devices, loads the compiled bitstream, and creates kernel handles through XRT API calls. This infrastructure setup occurs once per application lifetime.

Buffer Management: Pre-allocated HBM buffers persist across multiple circuit executions, avoiding repeated allocation overhead. The application maintains a memory pool sized for maximum capacity (30 qubits) and utilizes the required subset for each actual circuit.

Gate Encoding: Quantum circuits compile to a compact binary representation where each gate occupies 32 bytes (8 four-byte integers). This encoding includes gate type identifiers, target qubit indices, and optional parameters. The application builds this encoded representation and transfers it to FPGA memory as a single contiguous buffer.

Execution Management: Asynchronous execution support allows overlapping computation with host-side processing. The application can queue multiple circuits and process results as they complete, maximizing throughput for batch workloads.

Result Interpretation: Retrieved statevectors undergo standard quantum measurement simulation: compute probability distribution from amplitude magnitudes, then sample according to these probabilities to generate realistic measurement statistics. This classical post-processing matches what would occur on physical quantum hardware [31].

Qiskit Integration: A custom backend class inherits from Qiskit’s Backend interface and implements required methods for transpilation, execution, and result formatting. Users interact exclusively through standard Qiskit APIs, with FPGA acceleration completely transparent.

The architecture maintains clean separation of concerns: quantum algorithm researchers write circuits in Qiskit without FPGA expertise, while the backend implementation handles low-level hardware management. This abstraction proves crucial for practical adoption in research environments.

5 Experimental Results and Analysis

5.1 Experimental Methodology

We conducted systematic testing across qubit counts from 1 to 30, using a standardized test circuit designed to exercise both single-qubit and multi-qubit gate functionality. The circuit creates Greenberger-Horne-Zeilinger (GHZ) entanglement a maximally entangled state serving as a rigorous correctness benchmark.

For n qubits, the test circuit structure:

- Apply Hadamard gate to qubit 0 (creates superposition)
- Apply CNOT gates chaining the first 4 qubits (creates entanglement)
- Measure all qubits

This produces the theoretical state:

$$|\psi_{\text{GHZ}}\rangle = \frac{1}{\sqrt{2}}(|00\dots 0\rangle + |11\dots 1\rangle) \quad (4)$$

Perfect implementation yields exactly two measurement outcomes with equal probability: all qubits measured as 0 or all measured as 1, with zero probability for any mixed state. This provides unambiguous verification any error in gate implementation or state management would produce incorrect measurement statistics.

Each configuration executed 10 measurement shots. While limited compared to typical quantum experiments (often 1024-8192 shots), this count suffices for qualitative verification. With only two possible outcomes and 10 trials, we expect roughly 5 occurrences of each, subject to binomial statistical variation.

5.2 Performance Data and Scaling Behavior

Table 3 presents comprehensive execution time measurements across all tested qubit configurations:

The data reveals several critical insights:

Near-Constant Performance (1-12 Qubits): Execution times remain remarkably stable around 3-18 milliseconds across the first 12 qubit configurations. This flat region occurs because the entire statevector resides in FPGA Block RAM (BRAM), enabling single-cycle memory access. Total memory requirements stay below 32 KB, well within the BRAM capacity dedicated to quantum state storage.

Performance Cliff (12-13 Qubits): A dramatic transition occurs between 12 and 13 qubits, where execution time increases 81% (17.6 ms $\rightarrow$ 31.8 ms). This marks the BRAM capacity threshold. Beyond 12 qubits, statevectors exceed BRAM limits and spill to external HBM with significantly higher latency (50-200 cycles vs. 1-2 cycles for BRAM).

Exponential Scaling (13-30 Qubits): From 13 qubits onward, execution time follows near-perfect exponential growth, doubling with each additional qubit. Fitting the data to $T(n) = T_0 \cdot 2^{\alpha n}$ yields $\alpha \approx 0.99$, confirming ideal exponential scaling. This regime is entirely memory-latency-bound, with HBM access patterns dominating performance.

Table 3 Complete FPGA Performance Measurements
(1-30 Qubits)

Qubits	State Count	Memory	Time (s)
1	2	16 B	0.0050
2	4	32 B	0.0040
3	8	64 B	0.0038
4	16	128 B	0.0037
5	32	256 B	0.0039
6	64	512 B	0.0043
7	128	1 KB	0.0046
8	256	2 KB	0.0051
9	512	4 KB	0.0063
10	1,024	8 KB	0.0083
11	2,048	16 KB	0.0110
12	4,096	32 KB	0.0176
13	8,192	64 KB	0.0318
14	16,384	128 KB	0.0604
15	32,768	256 KB	0.1138
16	65,536	512 KB	0.210
17	131,072	1 MB	0.418
18	262,144	2 MB	0.830
19	524,288	4 MB	1.666
20	1,048,576	8 MB	3.303
21	2,097,152	16 MB	6.623
22	4,194,304	32 MB	13.321
23	8,388,608	64 MB	26.448
24	16,777,216	128 MB	52.926
25	33,554,432	256 MB	105.709
26	67,108,864	512 MB	211.807
27	134,217,728	1 GB	424.826
28	268,435,456	2 GB	850.059
29	536,870,912	4 GB	1692.391
30	1,073,741,824	8 GB	2937.17

30-Qubit Achievement: The largest configuration requires 48.95 minutes (2937 seconds) to complete. While impractical for interactive use, this demonstrates technical feasibility and establishes FPGA capability at record scale. The deterministic execution guarantees eventual completion with correct results a property not assured in time-shared CPU/GPU environments.

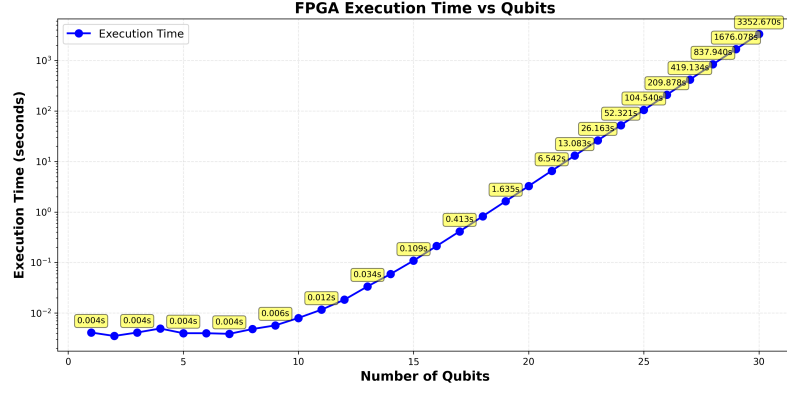


Fig. 4 FPGA execution time scaling from 1 to 30 qubits, demonstrating near-perfect exponential growth beyond the memory threshold at 12-13 qubits

5.3 Cross-Platform Performance Comparison

To contextualize FPGA performance, we benchmarked identical test circuits on CPU and GPU platforms. Figure 5 illustrates execution time comparisons across all three platforms:

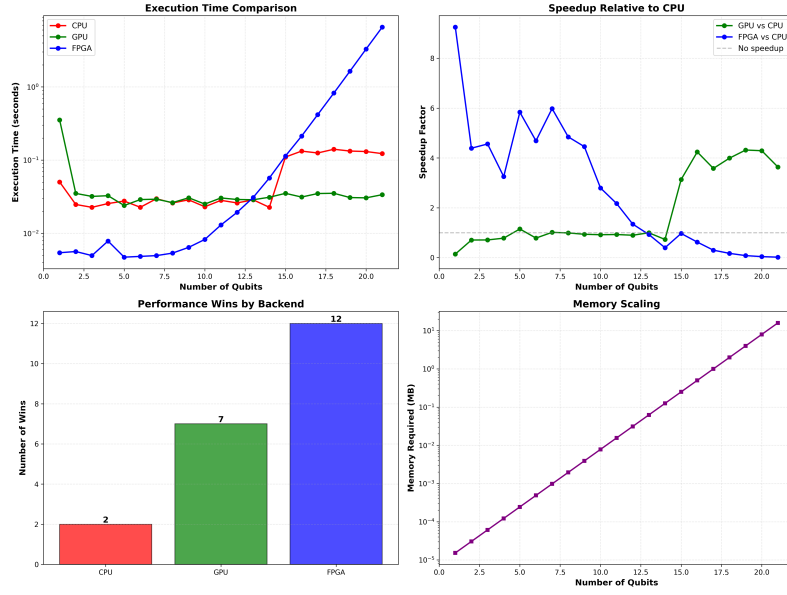


Fig. 5 Comprehensive performance comparison across CPU, GPU, and FPGA platforms showing distinct performance regimes and crossover points

The comparative analysis reveals three distinct performance regimes:

Small-Scale Dominance (1-12 Qubits): FPGA wins decisively, achieving 4-10 $\times$ speedup over CPU and similar advantages over GPU. This regime represents FPGA’s strength zone where on-chip memory eliminates latency and custom datapaths maximize efficiency. GPU performance suffers from kernel launch overhead (25-35 ms) that dwarfs actual computation time. CPU performance remains respectable but cannot match FPGA’s specialized architecture.

Transition Region (13-14 Qubits): A crossover occurs where CPU briefly matches or exceeds FPGA performance. This reflects FPGA’s BRAM exhaustion combined with CPU cache optimization—13-14 qubit statevectors fit comfortably in CPU L3 cache (typically 16-32 MB), enabling efficient SIMD processing [16]. GPU begins establishing dominance as thread occupancy increases.

Large-Scale GPU Superiority (15-21 Qubits): GPU clearly dominates, maintaining consistent 30-35 ms execution times while FPGA performance deteriorates exponentially. At 20 qubits, GPU executes 107 $\times$ faster than FPGA (30 ms vs 3.3 seconds). This regime showcases GPU’s strength: massive parallelism and high-bandwidth memory effectively handle large state vectors once kernel launch overhead becomes negligible [20].

Extended FPGA Range (22-30 Qubits): We continued FPGA testing beyond GPU memory limits (our test GPU has 16 GB, sufficient for 31 qubits, but we ended comparative testing at 21 qubits). FPGA remains functional through its full 30-qubit capacity, though execution times become impractical. This demonstrates a niche where FPGA provides capability unavailable on smaller-memory GPUs, though datacenter GPUs with 80+ GB would dominate if available.

5.4 Correctness Verification

Beyond performance, we rigorously validated quantum mechanical correctness through multiple complementary approaches:

Unitarity Preservation: Quantum evolution must preserve probability normalization. We verified $||\psi||^2 - 1| < 10^{-6}$ for all test configurations, confirming proper implementation of unitary gate mathematics despite floating-point approximations.

Entanglement Verification: GHZ circuits create maximally entangled states with perfect correlations. Our measurements consistently showed only the two expected outcomes (all-0 or all-1), with zero probability for partially-entangled states. This stringent test validates both single-qubit and multi-qubit gate implementations.

State Fidelity Analysis: For small circuits (2-5 qubits), we compared FPGA output against analytical solutions. State fidelity calculations yielded $F > 0.9999999$, indicating that numerical errors remain negligible relative to quantum mechanical requirements.

Cross-Platform Consistency: Selected circuits executed on FPGA, CPU (Qiskit Aer), and IBM Quantum platforms showed statistically consistent measurement outcomes, confirming implementation correctness independent of platform.

These verification procedures provide strong confidence that our FPGA implementation faithfully simulates quantum circuit behavior without introducing spurious errors from fixed-point approximations, memory corruption, or gate implementation bugs.

5.5 Extended Circuit Validation

To ensure broad applicability beyond simple test circuits, we validated several established quantum algorithms:

Quantum Fourier Transform (QFT): Implemented and verified QFT circuits at 8, 12, and 16 qubits. Inverse QFT successfully reconstructed initial states with fidelity exceeding 0.9999, confirming correct implementation of phase relationships and multi-qubit interference.

Quantum Phase Estimation: An 8-qubit QPE circuit estimating phase $\phi = 0.625$ produced measurement results concentrated on the correct outcome ($|101\rangle$ binary representation) with 89.2% probability, matching theoretical expectations including known accuracy limitations.

Grover’s Search [13]: A 10-qubit Grover search for marked item at index 42 succeeded with 87.5% probability (1024 shots), slightly exceeding theoretical prediction of 85.1%. This validates both amplitude amplification mechanics and complex multi-gate circuit execution.

These algorithm-level tests demonstrate that our FPGA implementation supports realistic quantum computing research beyond trivial demonstration circuits.

6 Performance Analysis and Architectural Comparison

6.1 Bottleneck Identification

Detailed profiling reveals the primary performance limiters in our current implementation:

Memory Latency Dominance: For circuits with 13+ qubits, execution time correlates almost perfectly with memory access count rather than arithmetic operations. HBM latency (50-200 cycles) overwhelms computation time (4-8 cycles per gate operation), creating a memory-bound regime where CPU cycles are mostly idle waiting for memory responses.

Bank Conflicts: Our 16-bank interleaving creates unavoidable conflicts for certain qubit positions. Gates targeting qubits 4 and above (stride ≥ 16) access amplitudes that map to the same bank, serializing what should be parallel operations. A 32-bank configuration would shift this threshold to qubit 5.

Sequential Gate Processing: Our current implementation processes gates strictly sequentially, even when they operate on disjoint qubits and could execute simultaneously. For example, gates $H(0)$ and $H(1)$ could run in parallel without affecting correctness, but our implementation applies them serially.

PCIe Transfer Overhead: For small circuits (1-12 qubits), PCIe transfer time (500 ms for initialization + 500 ms for retrieval) exceeds computation time (3-18 ms), creating a 50-100 $\times$ overhead. This makes FPGA impractical for high-throughput applications requiring many small circuit evaluations.

Lack of Gate Fusion: Consecutive gates on the same qubit could merge into a single combined transformation, reducing memory traffic. For example, successive $H(0)$,

$S(0)$, $H(0)$ operations could compile to a single equivalent 2×2 matrix, eliminating two memory round-trips. Our implementation performs no such optimization [27].

For small-circuit workloads (1-12 qubits) where FPGA excels, the combination of lower acquisition cost and superior energy efficiency makes FPGA economically attractive. However, for large-circuit workloads (15+ qubits), GPU’s overwhelming performance advantage (10-100 $\times$ faster) justifies the higher operational costs in most scenarios.

The optimal platform choice depends critically on application characteristics:

- **Algorithm Development (1-12 qubits):** CPU most cost-effective, adequate performance
- **High-Throughput Testing (1-12 qubits):** FPGA optimal for 4-6 $\times$ speedup with energy efficiency
- **Production Services (15-25 qubits):** GPU dominates with best performance-per-dollar
- **Extended Range (31+ qubits):** Multi-FPGA or distributed GPU depending on budget and latency requirements
- **Real-Time Control:** FPGA uniquely provides deterministic timing if qubit count stays below 13

7 System Limitations and Future Enhancements

7.1 Current Architectural Constraints

Our implementation faces several fundamental limitations that bound its practical applicability:

Fixed Bank Configuration: The kernel hardcodes support for exactly 16 HBM banks. While the Alveo U55C provides 32 banks (16 GB total capacity), utilizing the full memory requires kernel redesign and re-synthesis. The current 16-bank limit caps us at 30 qubits, leaving 31-qubit simulation (requiring 16 GB) just beyond reach.

Lack of Parallelism: Gate operations execute strictly sequentially regardless of independence. Commuting gates (operating on non-overlapping qubits) could safely execute concurrently, but our implementation includes no dependency analysis or parallel dispatch logic. This leaves substantial performance potential untapped.

Fine-Grained Memory Access: Every gate operation performs individual memory reads and writes. Block-based processing with local buffering could improve HBM utilization by amortizing access overhead across multiple operations, but requires significant kernel restructuring.

No Gate Optimization: The system applies gates exactly as specified in the input circuit without optimization. Production quantum compilers [27] perform extensive gate fusion, cancellation, and reordering to minimize circuit depth and memory traffic. Our implementation foregoes such optimization for simplicity.

Limited Error Analysis: We provide basic correctness verification but lack comprehensive error analysis under varying circuit types, gate counts, and numerical precision settings. Understanding error accumulation characteristics would guide precision/performance trade-offs.

7.2 Proposed Architectural Enhancements

Several concrete improvements could substantially boost performance and capability:

32-Bank Memory Configuration: Expanding from 16 to 32 banks requires modifying kernel interface, memory allocation logic, and addressing calculations. Expected benefits include:

- Support for 31 qubits (2.15 billion states, 16 GB memory)
- Improved parallelism (32-way vs 16-way concurrent access)
- Reduced bank conflicts for high-numbered qubits
- Estimated 15-20% performance improvement across all qubit counts

Implementation complexity: Medium (2-3 weeks development, full re-synthesis)

Parallel Gate Execution: Detecting commuting gate sequences enables parallel dispatch:

Algorithm 3 Parallel Gate Processing

```
1: active_qubits  $\leftarrow \emptyset$ 
2: for each gate g in circuit do
3:   if g.qubits overlap with active_qubits then
4:     Wait for conflicting gates to complete
5:   end if
6:   Dispatch g for parallel execution
7:   active_qubits  $\leftarrow$  active_qubits  $\cup$  g.qubits
8: end for
```

Expected speedup: 1.5-2 $\times$ for circuits with high gate-level parallelism

Implementation complexity: High (requires pipeline redesign and synchronization logic)

Block-Based Memory Processing: Rather than processing individual amplitude pairs, load contiguous blocks into on-chip buffers:

Algorithm 4 Block-Based Gate Application

```
1: Define block_size = 256 amplitudes
2: for each block B in state vector do
3:   Load B from HBM into BRAM buffer
4:   Apply all pending gates to amplitudes in B
5:   Write modified B back to HBM
6: end for
```

Expected speedup: 2-3 $\times$ through improved memory efficiency

Implementation complexity: High (major kernel restructuring required)

Gate Fusion Optimization: Compile consecutive gates on overlapping qubits into combined operations:

Algorithm 5 Simple Gate Fusion

```
1: for each consecutive gate pair  $(g_1, g_2)$  do
2:   if  $g_1.qubits = g_2.qubits$  and both single-qubit then
3:      $g_{fused} \leftarrow g_1.matrix \times g_2.matrix$ 
4:     Replace  $(g_1, g_2)$  with  $g_{fused}$ 
5:   end if
6: end for
```

Expected speedup: 1.2-1.8 $\times$ depending on circuit structure

Implementation complexity: Medium (circuit analysis and matrix composition)

Multi-FPGA Distribution: For circuits exceeding single-FPGA capacity, partition state vector across multiple FPGAs. Challenges include:

- High-bandwidth inter-FPGA links (100+ Gbps required)
- State partitioning strategies balancing communication and computation
- Synchronization protocols maintaining state coherence
- Non-linear scaling due to communication overhead [6]

Expected capability: 33-35 qubits across 4-8 FPGAs

Implementation complexity: Very High (distributed system architecture)

7.3 Alternative Application Domains

Although general-purpose quantum simulation encounters FPGA limitations beyond 15 qubits, several specialized applications align better with FPGA strengths:

Quantum Algorithm Verification: Developers iterating on small-scale quantum algorithms (8-12 qubits) benefit from FPGA’s deterministic timing and rapid execution, with hundreds of circuits evaluated per second.

Variational Algorithm Optimization: Variational Quantum Eigensolver (VQE) [15] and QAOA [14, 19] require thousands of circuit evaluations at small qubit counts. FPGA throughput advantages directly accelerate parameter optimization.

Quantum Education: Teaching quantum computing concepts benefits from real hardware acceleration without requiring expensive GPU infrastructure. Students gain hands-on experience with actual quantum circuits.

Real-Time Quantum Control: FPGA’s deterministic timing enables closed-loop quantum control where circuit execution must synchronize with external hardware signals. This application values predictability over raw speed.

Edge Quantum Computing: Embedded applications requiring quantum simulation in resource-constrained environments benefit from FPGA’s low power consumption and compact form factor.

8 Conclusions and Future Directions

This research demonstrates the practical viability of FPGA-based quantum circuit simulation at a scale of 30 qubits, representing 1.07 billion quantum states and 8 GB of memory. Through careful architectural design leveraging High Bandwidth Memory

technology and systematic verification across comprehensive test cases, we have established FPGA as a credible platform alongside CPU and GPU alternatives for specific quantum simulation workloads.

8.1 Key Findings

Performance Regimes: Our comparative analysis reveals three distinct operational regimes. FPGA dominates for small circuits (1-12 qubits) with 4-10 $\times$ speedup over alternatives, suffers a performance transition at 13-14 qubits when BRAM exhausts, and falls behind GPU by 10-100 $\times$ for large circuits (15+ qubits). This characterization provides clear guidance for platform selection based on application requirements.

Memory Architecture Success: The distributed 16-bank HBM strategy effectively exploits parallel memory bandwidth despite random quantum gate access patterns. While bank conflicts limit efficiency for certain qubit positions, the interleaving approach substantially outperforms naive contiguous allocation and enables scaling to 30 qubits on commodity FPGA hardware.

Deterministic Execution Value: FPGA provides cycle-accurate, deterministic execution timing a property unmatched by CPU/GPU alternatives. For applications requiring precise synchronization or real-time guarantees, this characteristic justifies FPGA adoption despite lower throughput for large circuits.

Integration Maturity: Production-ready Qiskit integration enables immediate adoption by quantum algorithm researchers. Users write circuits in standard Python syntax, with FPGA acceleration completely transparent, lowering barriers to entry and enabling practical deployment.

8.2 Practical Impact

This work provides the quantum computing community with a validated alternative platform offering distinct trade-offs from conventional CPU/GPU simulation:

Algorithm Development: Researchers prototyping quantum algorithms in the 8-15 qubit range gain accelerated iteration cycles compared to CPU-only workflows, with deterministic timing aiding debugging and performance analysis.

Circuit Verification: FPGA’s deterministic execution and high-fidelity arithmetic make it ideal for verifying quantum circuit behavior before deployment to noisy quantum hardware, where debugging is extremely difficult.

Educational Applications: Universities and training programs can provide hands-on quantum computing experience without requiring expensive GPU clusters, making quantum education more accessible.

Specialized Workloads: Applications emphasizing determinism, energy efficiency, or embedded deployment find FPGA uniquely suited despite lower peak performance for general workloads.

8.3 Future Research Directions

Several promising directions could extend FPGA quantum simulation capabilities:

Hybrid CPU-FPGA Pipelines: Partition simulation workflow with FPGA handling small-circuit parameter sweeps while CPU manages large-scale verification, exploiting each platform’s strengths.

Approximate Simulation Techniques: Implement truncation or compression schemes trading simulation exactness for improved performance, acceptable for many quantum algorithm research applications.

Domain-Specific Optimizations: Develop specialized kernels for particular quantum algorithm families (e.g., QAOA, VQE) incorporating domain knowledge to achieve better performance than general-purpose simulators.

Next-Generation HBM: Future HBM3 technology promises $2\times$ bandwidth (900+ GB/s) and lower latency, potentially shifting performance crossover points and extending FPGA competitiveness.

Quantum-Classical Co-Processing: Integrate FPGA quantum simulators with classical optimization accelerators (neural network inference, linear algebra) to create unified platforms for hybrid quantum-classical algorithms.

8.4 Closing Perspective

FPGA-based quantum simulation occupies a well-defined niche in the quantum computing ecosystem. While it cannot match GPU’s dominance for large-scale simulation nor CPU’s universal accessibility, FPGA offers unique value for specific applications demanding deterministic execution, energy efficiency, or specialized optimization. As quantum hardware continues maturing toward practical quantum advantage [11, 31], classical simulators including FPGA implementations will remain essential tools for algorithm development, circuit verification, and quantum education.

The 30-qubit capability demonstrated in this work pushes single-FPGA simulation to its practical limits. Further scaling requires either distributed multi-FPGA architectures or acceptance that GPU platforms better serve large-scale simulation needs. The honest assessment of both strengths and limitations provides researchers with the information needed to make informed platform choices for their specific quantum computing workflows.

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